NAME

hc_malloc – Sets up and manages memory allocation arena.

SYNOPSIS

hc_malloc [*options*] [*file*...]

DESCRIPTION

hc_malloc is a library written in C that sets up an information structure that keeps track of heap allocated memory allocations. The functions wrap around the standard memory allocation commands and keeps track of the addresses. It works seamlessly on Unix, Linux and macOS.

PUBLIC FUCTIONS

hc_init()

This function should be called fairly early in your program. It will set up an array that can contain up to 32 address pointer that can automatically be expanded in 32 additional pointer locations at the time when needed. Capacity and next available index values are part of this structure.

Returns:none

hc_cleanup()

You should call this functions at the end of your process. It will free all addresses in the address array. The memory allocated for the arena structure will be freed as well.

Returns:none

hc_free(void*addr_pntr)

Will free the memory allocated at the specified address. The address will be removed from the array.

Returns:none

hc_display()

Displays the content of the arena structure, including the address array.

Returns:none

hc_malloc(size_tsize)

Allocates given size memory and add it to the array and updates the next index value.

Returns:address**pointer**

hc_calloc(size_tnum_elem,**size_t**elem_size)

Allocates requested memory block. Size is calculated by multiplying the number of element requested by the size of the element. Address will be added to the arena array and updates the next index value.

Returns:address**pointer**

hc_realloc(void*old_pntr,**size_t**size)

Reallocates the existing memory alloction pointed to by the old_pntr argument to a new area with the specified size. The original address in the array will be updated with the new one.

Returns:address**pointer**

hc_strdup(void*addr_pntr)

Copies the content of a string pointed to by the addr_pntr. Updates the address array and next_index value.

Returns:char**address**pointer

INTERNAL FUCTIONS

_hc_address_array_capacity_increase()

This function will be invoked when the current capacity of the address array is reached. The increments is by 32 address locations each time.

Returns:none

_hc_address_add(addr_pntr)
Adds an address pointer to the address array. Will evoke _hc_address_array_capacity_increase() when the current capacity is reached

Returns:none

_hc_address_locate(void*addr_pntr)
Returns the index of the given address pointer.

Returns:int

_hc_address_replace(void*old_pntr,**void***new_pntr)
Locates the given old pointer in the address array and replaces it with the new given pointer.

Returns:none

EXAMPLES

FILES

/usr/local/share/hc_malloc/doc/README.md
Documentation to this utility, also available in html and pdf form.

LICENSE

/usr/local/share//hc_malloc/doc/LICENSE
MIT license.

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